

Section-A (2*5=10)

Q1. Differentiate NFA & DFA.

Ans. Deterministic Finite Automata

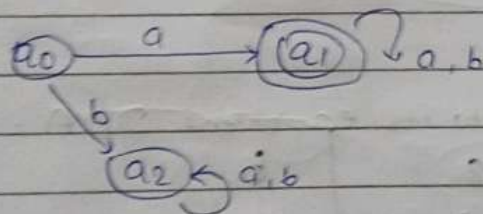
Non-deterministic Finite automata.

- $\delta: Q \times E \rightarrow Q$
- i.e., next possible state belongs to Q .
- Dead state allowed.
- ϵ move is not allowed.
- All DFA are NFA.
- Require more space.
- More difficult to construct.

- $\delta: Q \times E \rightarrow 2^Q$
- i.e., next possible state belongs to power set of Q .
- Dead state is not allowed.
- ϵ move is allowed.
- Not all NFA are DFA.
- Require less space.
- Easier to construct.

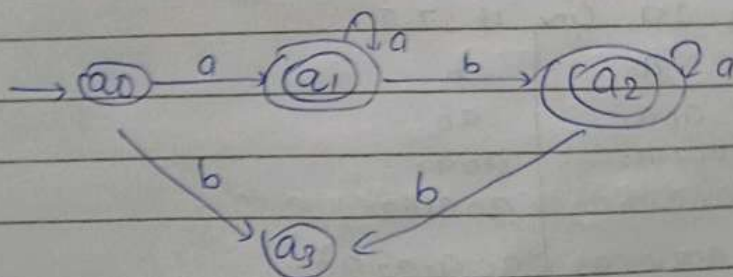
Q2. Draw a DFA for the language accepting string starting with 'a' over input alphabets $\Sigma = \{a, b\}$

Ans. $L = \{a, ab, aa, aba, aaa, abb, aab, \dots\}$
 State = $n+2 = 1+2 = 3$



Q3. Draw a DFA for the language accepting string start & end with 'a' over input alphabets $\Sigma = \{a, b\}$

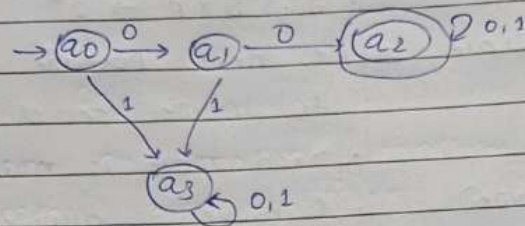
Ans. $L = \{a, aa, aba, aaa, \dots\}$



Q4. Draw a DFA that accepts a language L over input alphabet $\Sigma = \{0,1\}$ such that L is the set of all strings that end with '00'.

Ans $L = \{00, 001, 000, \dots\}$

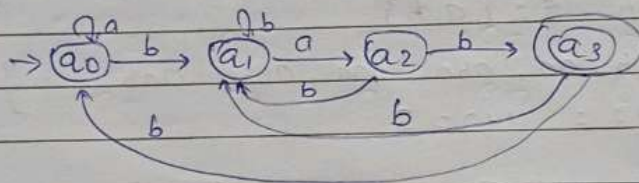
$$\text{States} = n+2 = 2+2 = 4$$



Q5. Draw a DFA for the language accepting strings ending with 'bab' over input alphabets $\Sigma = \{a,b\}$.

Ans $L = \{bab, abab, bbab, \dots\}$

$$\text{States} = n+1 = 3+1 = 4$$



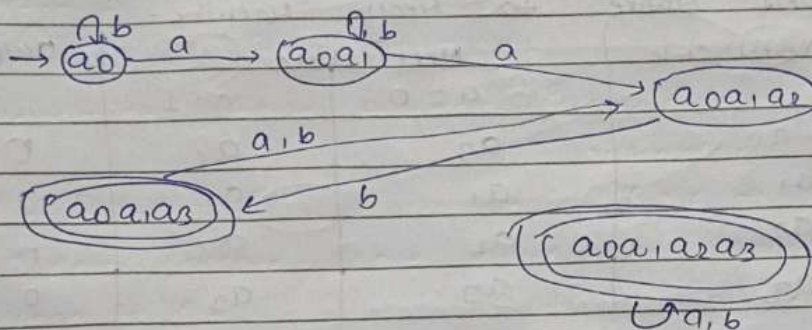
• Sec-B (7*3=21)

1. Convert the NFA into its equivalent DFA.

State/ Σ	a	b
$\rightarrow q_0$	$q_0 q_1$	q_0
q_1	q_2	q_1
q_2	q_3	q_3
(q_3)		q_2

Step ①:- Copy 1st line of T.T

②: Σ	a	b
$\rightarrow q_0$	$q_0 q_1$	q_0
$q_0 q_1$	$q_0 q_1 q_2$	$q_0 q_1$
$q_0 q_1 q_2$	$q_0 q_1 q_2 q_3$	$q_0 q_1 q_2$
$(q_0 q_1 q_2 q_3)$	$q_0 q_1 q_2 q_3$	$q_0 q_1 q_2 q_3$
$(q_0 q_1 q_2)$	$q_0 q_1 q_2$	$q_0 q_1 q_2$



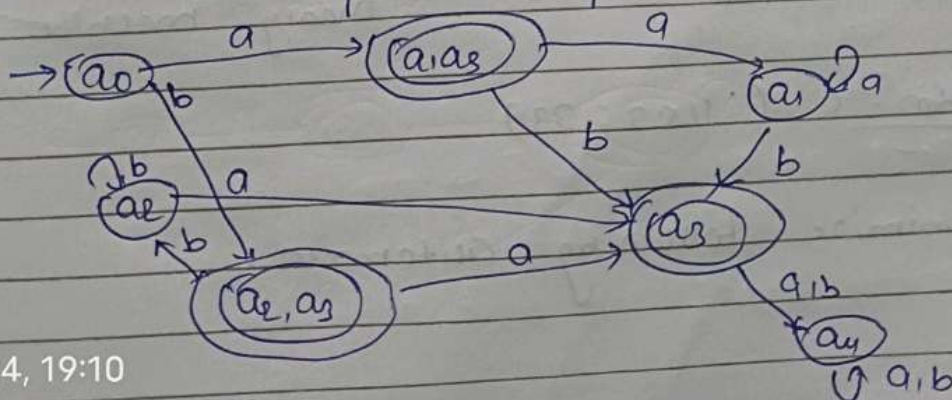
equivalent DFA.

2. Convert the NFA into its equivalent DFA.

State	a	b
q0	q1, q2	q2, q3
q1	q1	q3
q2	q3	q2
q3	-	-

Ans. copy the 1st line of T.T.

Σ	a	b
$\rightarrow q_0$	q1, q3	q2, q3
q1, q3	q1	q4
q1	q1	q3
q3	q4	q4
q2, q3	q3	q2
q2	q3	q2
q4	q4	q4



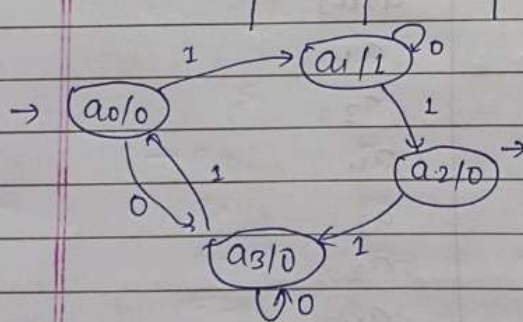
Q3. Convert Moore to Mealy Machine

Present state	Next state		output
	a = 0	a = 1	
- a ₀	a ₃	a ₁	0
a ₁	a ₁	a ₂	1
a ₂	a ₂	a ₃	0
a ₃	a ₃	a ₃	0

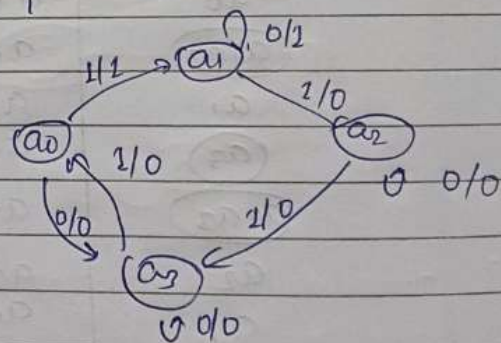
Ans ① Copy all moore machine transition state into mealy machine T.T

② Write out accordingly.

Σ	0	O/P	1	O/P
-a ₀	a ₃	0	a ₁	1
a ₁	a ₁	1	a ₂	0
a ₂	a ₂	0	a ₃	0
a ₃	a ₃	0	a ₀	0



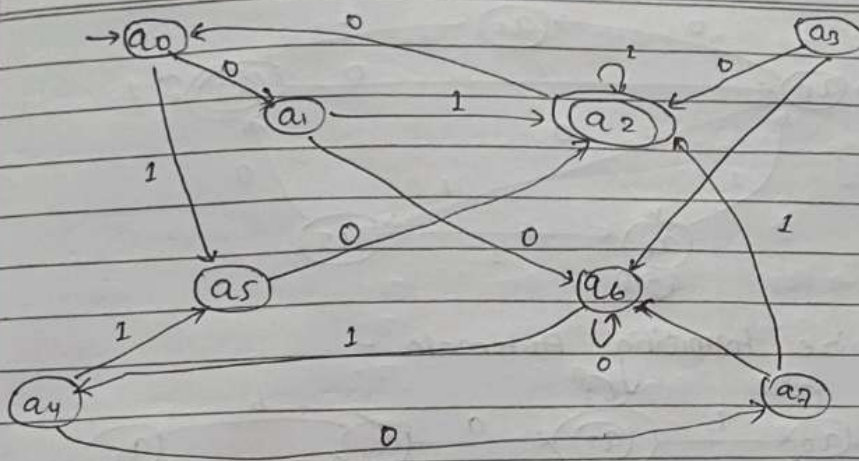
Moore Machine



mealy machine

• Section - C (11x3=33)

① minimize following automata:

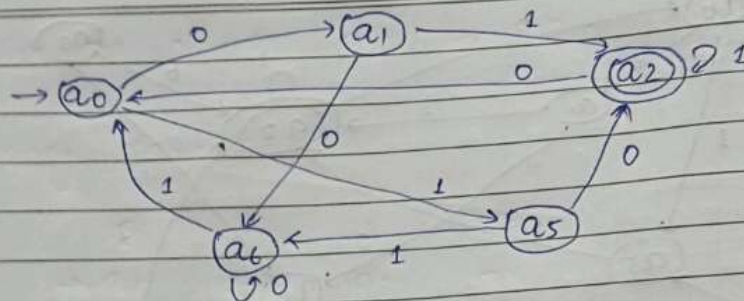


- solⁿ
- ① Eliminate unreachable state $\{q_3\}$
 - ② Separate final and non-final states.
 - ③ make TT & separate states that have same value for 0, 0 (non-final)

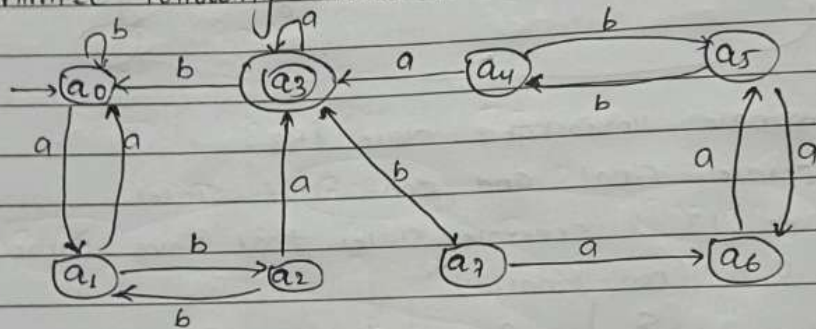
Σ	0	1
$\rightarrow q_0$	q_1	q_5
q_1	q_6	q_2
(q_2)	q_0	q_2
q_4	q_7	q_6
q_5	q_2	q_6
q_6	q_6	q_6
q_7	q_6	q_2

$\{q_0, q_4\}, \{q_1, q_7\}, \{q_5\}, \{q_6\}$ are new pairs of non-final
 $\{q_2\}$ is final.
 minimized T.T.

Σ	0	1
$\rightarrow q_0$	q_1	q_5
q_1	q_6	q_2
(q_2)	q_0	q_2
q_5	q_2	q_6
q_6	q_6	q_0



2. (32) Minimize following Automata -



Ans (i) eliminate unreachable state $\{q7\}$.

(ii) separate final & non-final states.

(iii) combine non-final states if they have same value for inputs.

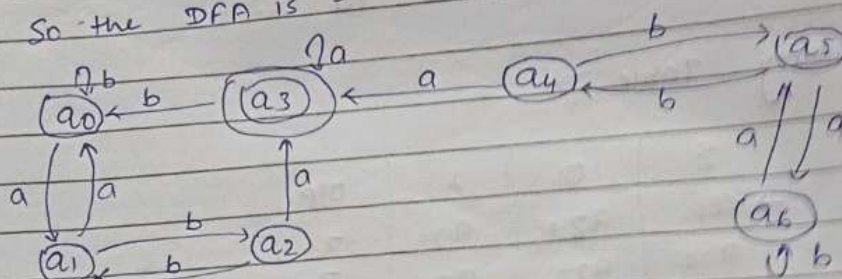
Σ	a	b
$\rightarrow q_0$	q_1	q_0
q_1	q_0	q_2
q_2	q_3	q_1
q_3	q_3	q_0
q_4	q_3	q_5
q_5	q_6	q_4
q_6	q_5	q_6

$\{q_0, q_1, q_2, q_4, q_5, q_6\} \rightarrow$ non-final

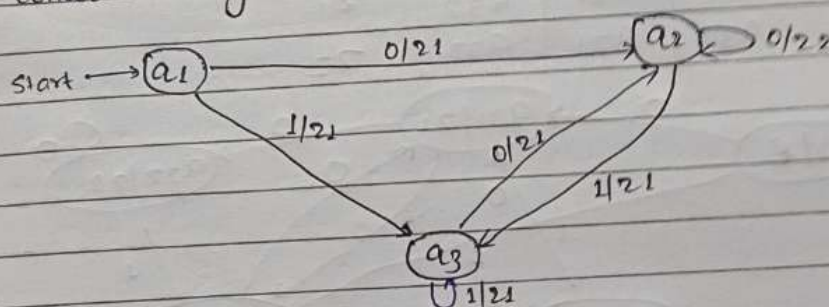
$\{q_3\} \rightarrow$ final

No non-final state have same value for inputs.

So the DFA is -



3. Convert Mealy to Moore machine.



Ans

- (i) Calculate no. of differ. o/p for each state that are available as mealy Table.
- (ii) if all output are same, copy state θ . if it has n distinct output break θ into n state.
- (iii) if output of initial state is 1, insert new initial state a beginning which gives 0 output.

Mealy state Table :-

Σ	0	o/p	1	o/p
$\rightarrow q_1$	q_2	z_1	q_3	z_1
q_2	q_2	z_2	q_3	z_1
q_3	q_2	z_1	q_3	z_2

From above table q_2 & q_3 has two output. So two state will form for both for each output i.e. q_{21}, q_{31}

for 21 & a22, a32, for 22.

Moorey Table :-

Σ	0	1	O/P
$\rightarrow a_1$	a_{21}	a_{31}	21
a_{21}	a_{22}	a_{31}	21
a_{22}	a_{22}	a_{31}	22
a_{31}	a_{21}	a_{32}	21
a_{32}	a_{21}	a_{32}	22

